

Shrey Goel

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EDUCATION

Duke University

Bachelor of Science, Computer Science & Mathematics (GPA: 3.83)

May 2027

Durham, NC

Related Coursework: Generative Models, Deep Learning, Natural Language Processing, Real Analysis I, Advanced Probability, Stochastic Processes, Linear Algebra, Multivariable Calculus, Data Structures & Algorithms, Databases

Dr. Bart Kamen Memorial Scholar: \$40,000 merit scholarship awarded to students with high research output

Volunteer Services: Workshop paper reviewer for the 2025 Neural Information Processing Systems Conference

PEER-REVIEWED ARTICLES

- Vincoff S., **Goel S.**, Kholina K., Pulugurta R., Vure P., & Chatterjee P. (2025). FusOn-pLM: a fusion oncoprotein-specific language model via adjusted rate masking. *Nature Communications*, 16(1), 1436.
- Smela M.P., Kramme C.C., Fortuna R.J.P., Wolf B., **Goel S.**, Adams J., Ma C., Velychko S., Widocki U., Kavirayuni V.S., Chen T., Vincoff S., Dong E., Kohman R.E., Kobayashi M., Shioda T., Church G.M., Chatterjee P. (2025). Rapid Human Oogonia-like Cell Specification via Combinatorial Transcription Factor-Directed Differentiation. *EMBO Reports*, 1-30.
- Bhat S., Palepu K., ..., **Goel S.**, ... & Chatterjee, P. (2025). De novo design of peptide binders to conformationally diverse targets with contrastive language modeling. *Science Advances*, 11(4), eadr8638.
- Chen T., Dumas M., Watson R., Vincoff S., Peng C., Zhao L., Hong L., Pertsemliadis S., Shaepers-Cheu M., Wang T., Srijay D., Monticello C., Vure P., Pulugurta R., Kholina K., **Goel S.**,... & Chatterjee, P. (2024). Target sequence-conditioned design of peptide binders using masked language modeling. *Nature Biotechnology*, 1-9.

PREPRINTS

- Goel S.**, Chatterjee P. (2026). Minimal Action Discrete Schrödinger Bridge Matching. *arXiv preprint arXiv:2601.22408*.
- Goel S.**, Schray P., Zhang Y., Vincoff S., Kratochvil H., Chatterjee P. (2025). Token-Level Guided Discrete Diffusion for Membrane Protein Design. *arXiv preprint arXiv:2410.16735*.
- Hong L., Ye T., Wang T., Srijay D., Zhao L., Watson R., Vincoff S., Chen T., Kholina K., **Goel S.**,... & Chatterjee P. (2024). Programmable Protein Stabilization with Language Model-Derived Peptide Guides. *Research Square*, rs-3.
- Ye T., Alamgir A., Robertus C., Colina D., Monticello C., Donahue T.C., Hong L., Vincoff S., **Goel S.**,... & DeLisa MP. (2024). Programmable protein degraders enable selective knockdown of pathogenic β -catenin subpopulations in vitro and in vivo. *bioRxiv*, 2024-11.

PROCEEDINGS

- Goel S.**, Chatterjee P. (2026). Minimal Action Discrete Schrödinger Bridge Matching. *International Conference on Learning Representations – ReALM-GEN Workshop for Generative Models*.
- Goel S.**, Schray P., Zhang Y., Vincoff S., Kratochvil H., Chatterjee P. (2025). Token-Level Guided Discrete Diffusion for Membrane Protein Design. *Neural Information Processing Systems – AI4Science Workshop (Oral Presentation)*.
- Goel S.**, Thoutam V., Marroquin E. M., Gokaslan A., Firouzbakht A., Vincoff S., ... & Chatterjee P. (2025). MeMDLM: De Novo Membrane Protein Design with Property-Guided Discrete Diffusion. *International Conference on Learning Representations – Generative Perspectives for Biology Workshop*.

RESEARCH EXPERIENCE

Machine Learning Research @ Coley Lab

Project #1

- Geometric generative modeling for 3D macrocycle design.

May 2026 – Present

Massachusetts Institute of Technology

Machine Learning Research @ Chatterjee Lab

Project #1

- Formulated MadSBM, a theoretical generative modeling framework for discrete biological sequences using Schrödinger Bridge Matching theory.
- Derived a simulation-free loss function to learn optimal Schrödinger bridge control fields, avoiding costly forward-backward iterative projections.

April 2023 – Present

University of Pennsylvania

- Implemented and trained Diffusion Transformer (DiT)-based model in PyTorch, demonstrating MadSBM models remain in high-probability regions compared to state-of-the-art diffusion and flow-matching models.
- **First-author** manuscript accepted at ICLR 2026 (generative models workshop).

Project #2

- Developed and trained discrete diffusion model for membrane protein sequence generation on High-Performance Computing cluster using PyTorch Lightning, Wandb, and HuggingFace.
- Generated protein sequences achieved wet-lab performance equivalent to naturally existing controls and a 44% decrease in perplexity over state-of-the-art autoregressive models.
- Designed and implemented novel classifier-guided sampling algorithm combining attention scores and classifier gradients to selectively edit specific sequence tokens during inference.
- **First-author manuscript** published at **NeurIPS 2025** AI4Science workshops and selected for **oral presentation** .

Project #3

- Fine-tuned ESM-2 protein language model on a novel masked language modeling objective that leverages dynamic masking rates to engineer cancer protein-specific sequence embeddings.
- Trained over 40 model variants to reach a lower perplexity compared to standard MLM fine-tuning, conducting extensive ablation studies and hyperparameter tuning.
- Evaluated embedding quality by training Scikit-learn classifiers on downstream tasks, achieving a 25% improvement in AUROC and F1 score over pretrained embeddings.
- **Second-author manuscript** published at **ICML 2024** workshops and published in *Nature Communications* .

Gameto

April 2023 – Feb 2025

Bioinformatics Researcher

Duke University

- Reduced experimental analysis time from 4 months to 3 days by training language models for cell type classification.
- Created automated statistical analysis pipeline in R and Python to plot PCAs, volcano plots, and heat maps.
- Co-author manuscript published in *EMBO Reports*

PROFESSIONAL EXPERIENCE

Latus Bio

September 2025 – Present

Machine Learning Engineer (Part-Time)

Remote

- Training sequence classifiers and multi-objective discrete flow matching models for therapeutic AAV design

Qualcomm Technologies

May 2025 – August 2025

Machine Learning Engineer Intern

San Diego, CA

- Diagnosed performance bottlenecks in Meta's Llama LLM using PyTorch Profiler, identifying redundant computations over padded tokens in KV-cache system.
- Engineered optimized self-attention tensor computations that bypass pad tokens, reducing inference latency by 14x in quantized Llama models deployed on edge-devices.

TECHNICAL SKILLS

Languages: Python, Java, R

ML Technologies: PyTorch, Hugging Face, Scikit-learn, Parameter Efficient Fine Tuning, HPC, Pandas, NumPy

Developer Tools: Git, Jira, Docker, Jupyter, Figma